

$\text{tt_true}(\text{expr}('P')) \rightarrow \text{False} \rightarrow \text{Dependent on if P is true/false}$
 $\text{tt_true}(\text{expr}(' \sim P')) \rightarrow \text{False} \rightarrow \text{Only true if P is false}$
 $\text{tt_true}(\text{expr}('P \mid \sim P')) \rightarrow \text{True} \rightarrow \text{True if P or not P}$
 $\text{tt_entails}(\text{expr}('P \& Q'), \text{expr}('P')) \rightarrow \text{True} \rightarrow \text{P and Q are true therefore P is true}$
 $\text{tt_entails}(\text{expr}('P \& Q'), \text{expr}(' \sim(\sim P \mid \sim Q)')) \rightarrow \text{True} \rightarrow \text{P and Q are true therefore P and Q are true}$
 $\text{tt_entails}(\text{expr}('P \& Q'), \text{expr}('P')) \rightarrow \text{True} \rightarrow \text{P and Q are true, therefore P is true}$
 $\text{tt_entails}(\text{expr}('P \& Q'), \text{expr}(' \sim P')) \rightarrow \text{False} \rightarrow \text{P and Q are true, therefore not P is false}$
 $\text{dpll_satisfiable}(\text{expr}('P \& Q')) \rightarrow \{Q \rightarrow \text{True} \mid P \rightarrow \text{True}\} \rightarrow \text{P and Q can both be true}$
 $\text{dpll_satisfiable}(\text{expr}('P \& \sim Q')) \rightarrow \{Q \rightarrow \text{False} \mid P \rightarrow \text{True}\} \rightarrow \text{P can be true and Q can be false}$
 $\text{dpll_satisfiable}(\text{expr}('P \& Q \& \sim P')) \rightarrow \text{False} \rightarrow \text{P and Q and not P is false as } P \neq \text{not } P$

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Python Console>>> from logic import *
>>> tt_true(expr('P'))
False
>>> tt_true(expr(' ~P'))
False
>>> tt_true(expr('P | ~P'))
True
>>> tt_true(expr('P & ~P'))
False
>>> tt_entails(expr('P & Q'), expr('P'))
True
>>> tt_entails(expr('P & Q'), expr(' ~(~P | ~Q)'))
True
>>> tt_entails(expr('P & Q'), expr('P'))
True
>>> tt_entails(expr('P & Q'), expr(' ~P'))
False
>>> dpll_satisfiable(expr('P & Q'))
{Q: True, P: True}
>>> dpll_satisfiable(expr('P & ~Q'))
{Q: False, P: True}
>>> dpll_satisfiable(expr('P & Q & ~P'))
False
>>>

```